

Supplementary raw data on cell chaining in an attenuated *Listeria monocytogenes* mutant

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Abstract

The dataset “Supplementary raw data on cell chaining in an attenuated *Listeria monocytogenes* mutant” provides the raw images and data for the manuscript “On the role of cell chaining in the attenuation of a *Listeria monocytogenes* divIVA mutant” (<https://doi.org/10.64898/2026.01.05.697648>).

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Information on the dataset and context of origin

The dataset “Supplementary raw data on cell chaining in an attenuated *Listeria monocytogenes* mutant” provides the raw images and data, which form the basis for the figures of the manuscript “On the role of cell chaining in the attenuation of a *Listeria monocytogenes* divIVA mutant”. In-depth information on the background of the work, on methodology and on the analysis of the results can be found in

Wamp, S., Holland, G., Rismondo, J., and Halbedel, S. (2026). On the role of cell chaining in the attenuation of a *Listeria monocytogenes* divIVA mutant. bioRxiv.
<https://doi.org/10.64898/2026.01.05.697648>

Administrative and organizational information

The publication “Supplementary raw data on cell chaining in an attenuated *Listeria monocytogenes* mutant”, is a joined work of the Consultant Laboratory for *Listeria* in [Unit 11 | Enteropathogenic Bacteria and Legionella](#) of the Robert Koch Institute, the Unit [ZBS 4 | Advanced Light and Electron Microscopy](#) and the Institute for [Medical Microbiology and Hospital Hygiene](#) at the Otto von Guericke University. Questions regarding the content of the data can be addressed directly to the corresponding author Sven Halbedel (halbedels@rki.de).

The publication of the data as well as the quality management of the (meta-)data is done by the unit [MF 4 | Domain Specific Data and Research Data Management](#). Questions regarding data management and the publication infrastructure can be directed to the Open Data Team of the unit MF4 at OpenData@rki.de.

Overview over the *L. monocytogenes* strains used for the manuscript

The following strains were used for the manuscript. Detailed methodology on their generation can be found in the methods section of the manuscript (<https://doi.org/10.64898/2026.01.05.697648>).

Strain name	Genetic background	Description
EGD-e	wt	Wildtype strain <i>L. monocytogenes</i> , used as control
LMSW49	<i>imreB</i>	MreB depletion strain, mreB expression can be induced upon IPTG-treatment
LMJR183	<i>iezrA</i>	EzrA depletion strain, ezrA expression can be induced upon IPTG-treatment
LMS2	Δ <i>divIVA</i>	divIVA deletion strain

Strain name	Genetic background	Description
LMSW207	$\Delta divIVA$ <i>imreB</i>	<i>divIVA</i> deletion and MreB depletion strain, mreB expression can be induced upon IPTG-treatment
LMSW208	$\Delta divIVA$ <i>iezrA</i>	<i>divIVA</i> deletion and EzrA depletion strain, ezrA expression can be induced upon IPTG-treatment
BUG2214	$\Delta prfA$	<i>prfA</i> deletion strain
LMS30	<i>idivIVA</i>	<i>DivIVA</i> depletion strain, <i>divIVA</i> expression can be induced upon IPTG-treatment
LMJD20	wt	DsRed producing <i>L. monocytogenes</i> strain, used as control
LMSF2	$\Delta actA$	DsRed producing, <i>actA</i> deletion strain
LMSW205	$\Delta divIVA$	DsRed producing, <i>divIVA</i> deletion strain
LMSW202	<i>iezrA</i>	DsRed producing, EzrA depletion strain, ezrA expression can be induced upon IPTG-treatment
LMSW203	<i>imreB</i>	DsRed producing, MreB depletion strain, mreB expression can be induced upon IPTG-treatment
BUG3057	<i>prfA</i> *	<i>L. monocytogenes</i> strain that encodes a PrfA variant that is constitutively active causing constitutive overexpression of PrfA-dependent virulence genes
LMSW251	$\Delta divIVA$ <i>prfA</i> *	<i>divIVA</i> deletion strain that encodes a PrfA variant that is constitutively active causing constitutive overexpression of PrfA-dependent virulence genes
LMSW222	$\Delta divIVA$ <i>secA2</i> M95I	<i>divIVA</i> deletion strain with <i>secA2</i> mutation (M95I)
LMSW223	$\Delta divIVA$ <i>secA2</i> T123A	<i>divIVA</i> deletion strain with <i>secA2</i> mutation (T123A)
Δiap	$\Delta cwhA$	p60 (CwhA) deletion strain
LMSW268	$\Delta divIVA$ <i>secA2</i> M95I <i>prfA</i> *	<i>divIVA</i> deletion strain with <i>secA2</i> mutation (M95I) that additionally encodes a PrfA variant that is constitutively active causing

Strain name	Genetic background	Description
		constitutive overexpression of PrfA-dependent virulence genes
LMSW269	$\Delta divIVA$ <i>secA2 T123A prfA*</i>	<i>divIVA</i> deletion strain with <i>secA2</i> mutation (T123A) that additionally encodes a PrfA variant that is constitutively active causing constitutive overexpression of PrfA-dependent virulence genes

Content and structure of the raw supplemented raw images and data

The raw data and relevant calculations that form the basis for the manuscript's figures are provided as .xlsx files.

The raw microscopy images, as well as uncropped images from a western blot, a gel electrophoresis and culture plates are provided in the "Images" subfolder as .tif files.

[Images/](#)

The naming and content of the image and data files are explained in the tables below per figure.

Figure 1

Figure	Description	File
Figure 1A	Growth curves of the strains EGD-e (wt) and LMSW49 (<i>imreB</i>) $\pm$ IPTG	Figure_1A_raw_data.xlsx
Figure 1B	Uncropped Western blot showing MreB and DivIVA expression levels in the strains EGD-e (wt) and LMSW49 (<i>imreB</i>) $\pm$ IPTG	Images/Figure_1B_uncropped_blot_images.tif
Figure 1C	Fluorescence microscopy image of EGD-e (wt, Nile red)	Images/Figure_1C_wt_raw_image.tif
Figure 1C	Fluorescence microscopy image of LMSW49 (<i>imreB</i>) with IPTG (Nile red)	Images/Figure_1C_imreB+IPTG_raw_image.tif
Figure 1C	Fluorescence microscopy image of LMSW49 (<i>imreB</i>) without IPTG (Nile red)	Images/Figure_1C_imreB-IPTG_raw_image.tif

Figure 2

Figure	Description	File
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Figure 2B	Microscopy image of EGD-e (wt) in phase contrast	Images/Figure_2B_wt_pc_raw_image.tif
Figure 2B	Fluorescence microscopy image of <i>iezrA</i> without IPTG (Nile red)	Images/Figure_2B_iezrA-IPTG_nile_red_raw_image.tif
Figure 2B	Microscopy image of <i>iezrA</i> without IPTG in phase contrast	Images/Figure_2B_iezrA-IPTG_pc_raw_image.tif
Figure 2B	Fluorescence microscopy image of <i>iezrA</i> with IPTG (Nile red)	Images/Figure_2B_iezrA+IPTG_nile_red_raw_image.tif
Figure 2B	Microscopy image of <i>iezrA</i> with IPTG in phase contrast	Images/Figure_2B_iezrA+IPTG_pc_raw_image.tif
Figure 2C	Average cell length of the strains EGD-e (wt) and LMJR183 (<i>iezrA</i>) $\pm$ IPTG	Figure_2C_raw_data.xlsx

Figure 3

Figure	Description	File
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Figure 3B	Fluorescence microscopy image of EGD-e (wt, Nile red)	Images/Figure_3B_wt_raw_image.tif
Figure 3B	Fluorescence microscopy image of Δ <i>divIVA</i> (LMS2, Nile red)	Images/Figure_3B_divIVA_raw_image.tif
Figure 3C	Number of cells per chain of the strains EGD-e (wt) and LMS2 (Δ <i>divIVA</i>)	Figure_3C_raw_data.xlsx

Figure 4

Figure	Description	File
Figure 4	Scanning electron microscopy image of wt	Images/Figure_4_wt_raw_image.tif
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Figure 4	Scanning electron microscopy image of LMSW49 (<i>imrB</i>) without IPTG	Images/Figure_4_mreB_image.tif

Figure	Description	File
Figure 4	Scanning electron microscopy image of LMJR183 (<i>iezrA</i>) without IPTG	Images/Figure_4_ezrA_image.tif

Figure 5

Figure	Description	File
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Figure 5A	Fluorescence microscopy image of $\Delta divIVA$ <i>iezrA</i> (LMSW208, Nile red) without IPTG	Images/Figure_5A_divIVA_ezrA_raw_image.tif
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Figure 5B	Scanning electron microscopy image of LMSW208 ($\Delta divIVA$ <i>iezrA</i>) without IPTG	Images/Figure_5B_divIVA_ezrA_raw_image.tif

Figure 6

Figure	Description	File
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Figure 6C	Plate photographs of plaque formation assays with EGD-e (wt), LMS30 (<i>idivIVA</i>), LMSW49 (<i>imreB</i>) and LMJR183 (<i>iezrA</i>) $\pm$ IPTG	Images/Figure_6C_raw_image.tif
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Figure 6D	Fluorescence microscopy image of macrophages infected with LMSF2 ($\Delta actA$)	Images/Figure_6D_actA_raw_image.tif
Figure 6D	Fluorescence microscopy image of macrophages infected with LMSW205 ($\Delta divIVA$)	Images/Figure_6D_divIVA_raw_image.tif

Figure	Description	File
Figure 6D	Fluorence microscopy image of macrophages infected with LMSW202 (<i>iezrA</i>) without IPTG	Images/Figure_6D_ezrA_raw_image.tif
Figure 6D	Fluorence microscopy image of macrophages infected with LMSW203 (<i>imreB</i>) without IPTG	Images/Figure_6D_mreB_raw_image.tif

Figure 7

Figure	Description	File
Figure 7A	(Relative) invasion rates into hepatocytes by EGD-e (wt), BUG2214 ($\Delta prfA$), BUG3057 (<i>prfA</i> [*]), LMS2 ($\Delta divIVA$) and LMSW251 ($\Delta divIVA prfA$ [*])	Figure_7A_raw_data.xlsx
Figure 7B	Intracellular growth in macrophages of EGD-e (wt), BUG2214 ($\Delta prfA$), BUG3057 (<i>prfA</i> [*]), LMS2 ($\Delta divIVA$) and LMSW251 ($\Delta divIVA prfA$ [*])	Figure_7B_raw_data.xlsx
Figure 7C	Plate photographs of plaque formation assays with EGD-e (wt), BUG2214 ($\Delta prfA$), BUG3057 (<i>prfA</i> [*]), LMS2 ($\Delta divIVA$) and LMSW251 ($\Delta divIVA prfA$ [*])	Images/Figure_7C_raw_image.tif

Figure 8

Figure	Description	File
Figure 8B	Fluorescence microscopy image of EGD-e (wt, Nile red)	Images/Figure_8B_wt_raw_image.tif
Figure 8B	Fluorescence microscopy image of LMS2 ($\Delta divIVA$, Nile red)	Images/Figure_8B_divIVA_raw_image.tif
Figure 8B	Fluorescence microscopy image of LMSW222 ($\Delta divIVA secA2 M95I$, Nile red)	Images/Figure_8B_divIVA_secA2_M95I_raw_image.tif
Figure 8B	Fluorescence microscopy image of LMSW223 ($\Delta divIVA secA2 T123A$, Nile red)	Images/Figure_8B_divIVA_secA2_T123A_raw_image.tif
Figure 8C	SDS-PAGE image of EGD-e (wt), LMS2 ($\Delta divIVA$), LMSW222 ($\Delta divIVA secA2 M95I$), LMSW223 ($\Delta divIVA secA2 T123A$) and Δiap ($\Delta cwhA$)	Images/Figure_8C_uncropped_gel_image_revised.tif
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Figure	Description	File
	and LMSW223 (<i>ΔdivIVA secA2 T123A</i>)	
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Figure 8G	Plate photographs of plaque formation assays with EGD-e (wt), LMS2 (<i>ΔdivIVA</i>), LMSW222 (<i>ΔdivIVA secA2 M95I</i>), LMSW223 (<i>ΔdivIVA secA2 T123A</i>), LMSW251 (<i>ΔdivIVA prfA*</i>), LMSW268 (<i>ΔdivIVA secA2 M95I prfA*</i>) and LMSW269 (<i>ΔdivIVA secA2 M95I T123A prfA*</i>)	Images/Figure_8G_raw_image.tif

Supplementary Figure S1

Figure	Description	File
Figure S1	Scanning electron microscopy image of EGD-e (wt)	Images/Figure_S1_wt_raw_image.tif
Figure S1	Scanning electron microscopy image of LMS2 (<i>ΔdivIVA</i>)	Images/Figure_S1_divIVA_raw_image.tif
Figure S1	Scanning electron microscopy image of LMJR183 (<i>iezrA</i>)	Images/Figure_S1_ezrA_raw_image.tif
Figure S1	Scanning electron microscopy image of LMSW49 (<i>imreB</i>)	Images/Figure_S1_mreB_raw_image.tif

Supplementary Figure S2

Figure	Description	File
Figure S2	Growth curves of the strains LMSW207 (<i>ΔdivIVA imreB</i>) and LMSW208 (<i>ΔdivIVA iezrA</i>)	Figure_S2_raw_data.xlsx

Supplementary Figure S3

Figure	Description	File
Figure S3	Fluorescence microscopy image of macrophages infected with LMJD20 (wt, DAPI)	Images/Figure_S3_wt_DAPI_raw_image.tif

Figure	Description	File
Figure S3	Fluorescence microscopy image of macrophages infected with LMJD20 (wt, phase contrast)	Images/Figure_S3_wt_pc_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMJD20 (wt, TRITC)	Images/Figure_S3_wt_TRITC_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW202 (<i>iezrA</i> , DAPI) without IPTG	Images/Figure_S3_ezrA_DAPI_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW202 (<i>iezrA</i> , phase contrast) without IPTG	Images/Figure_S3_ezrA_pc_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW202 (<i>iezrA</i> , TRITC) without IPTG	Images/Figure_S3_ezrA_TRITC_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW203 (<i>imreB</i> , DAPI) without IPTG	Images/Figure_S3_mreB_DAPI_raw_image.tif
Figure S3	Fluorescence microscopy image of macrophages infected with LMSW203 (<i>imreB</i> , phase contrast) without IPTG	Images/Figure_S3_mreB_pc_raw_image.tif
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Figure S3	Fluorescence microscopy image of macrophages infected with LMSW205 (Δ <i>divIVA</i> , DAPI)	Images/Figure_S3_divIVA_DAPI_raw_image.tif
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Supplementary Figure S4

Figure	Description	File
Figure S4	Phospholipolytic activity on egg-yolk agar of EGD-e (wt)	Images/Figure_S4_egg_yolk_agar_wt_raw_image.tif

Figure	Description	File
Figure S4	Phospholipolytic activity on egg-yolk agar of LMS2 ($\Delta divIVA$)	Images/Figure_S4_egg_yolk_agar_divIVA_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of BUG2214 ($\Delta prfA$)	Images/Figure_S4_egg_yolk_agar_prfA_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of BUG3057 ($prfA^*$)	Images/Figure_S4_egg_yolk_agar_prfA-star_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of LMSW251 ($\Delta divIVA prfA^*$)	Images/Figure_S4_egg_yolk_agar_divIVA-prfA-star_raw_image.tif
Figure S4	Hemolytic activity on sheep blood agar of EGD-e (wt), BUG2214 ($\Delta prfA$) and BUG3057 ($prfA^*$)	Images/Figure_S4_hemolysis_wt_prfA_prfA-star_raw_image.tif
Figure S4	Hemolytic activity on sheep blood agar of LMS2 ($\Delta divIVA$) and LMSW251 ($\Delta divIVA prfA^*$)	Images/Figure_S4_hemolysis_divIVA_divIVA-prfA-star_raw_image.tif
Figure S4	Swarming on soft agar of EGD-e (wt), BUG2214 ($\Delta prfA$), BUG3057 ($prfA^*$), LMS2 ($\Delta divIVA$) and LMSW251 ($\Delta divIVA prfA^*$)	Images/Figure_S4_swarming_plate_raw_image.tif
Figure S4	Cellular morphology after Nile red staining of EGD-e (wt)	Images/Figure_S4_nile_red_wt_raw_image.tif
Figure S4	Cellular morphology after Nile red staining of LMS2 ($\Delta divIVA$)	Images/Figure_S4_nile_red_divIVA_raw_image.tif
Figure S4	Cellular morphology after Nile red staining of BUG2214 ($\Delta prfA$)	Images/Figure_S4_nile_red_prfA_raw_image.tif
Figure S4	Cellular morphology after Nile red staining of BUG3057 ($prfA^*$)	Images/Figure_S4_nile_red_prfA-star_raw_image.tif
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Metadata

To increase findability, the provided data are described with metadata. The Metadata are distributed to the relevant platforms via GitHub Actions. There is a specific metadata file for each platform; these are stored in the metadata folder:

[Metadata/](#)

Versioning and DOI assignment are performed via [Zenodo.org](#). The metadata prepared for import into Zenodo are stored in the [zenodo.json](#). Documentation of the individual metadata variables can be found at <https://developers.zenodo.org/#representation>.

[Metadata/zenodo.json](#)

The zenodo.json includes the publication date and the date of the data status in the following format (example):

```
"publication_date": "2024-06-19",
"dates": [
  {
    "start": "2023-09-11T15:00:21+02:00",
    "end": "2023-09-11T15:00:21+02:00",
    "type": "Created",
    "description": "Date when the published data was created"
  }
],
```

Guidelines for reuse of the data

Open data from the RKI are available on [Zenodo.org](https://zenodo.org), [GitHub.com](https://github.com), [OpenCoDE](https://opencode.de), and [Edoc.rki.de](https://edoc.rki.de):

- <https://zenodo.org/communities/robertkochinstitut>
- <https://github.com/robert-koch-institut>
- <https://gitlab.opencode.de/robert-koch-institut>
- <https://edoc.rki.de/>

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